

03.2 Cloud Shell

Cloud Shell

- **Overview:**
 - Cloud Shell is an online development and operations environment provided by Google Cloud Platform.
 - It is accessible anywhere with a browser.
- **Features:**
 - Provides command-line access to a virtual machine instance in a terminal window.
 - Comes with favorite command-line tools pre-installed, up-to-date, and ready to use:
 - Bash, sh, emacs, vim
 - gcloud, MySQL, Kubernetes CLI, Docker CLI, minikube, and many more.
- **Storage:**
 - Includes 5 GB of persistent disk storage for the \$HOME directory.
- **Additional Tools:**
 - Online Code Editor: Develop, build, debug, and test your apps anywhere using the Cloud Shell Editor.
 - Source control via Git: Clone or pull remote repositories and commit code changes back to your repo using Cloud Shell's git client.

Cloud SDK - gcloud Command-line Tool

- **Overview:**
 - The gcloud command-line tool is part of the Google Cloud SDK.
 - It allows management of many Google Cloud Platform (GCP) services through the command line.
- **Services Managed by gcloud CLI:**
 - Google Compute Engine VMs and other resources.
 - Cloud SQL Instances, Cloud Functions, Cloud Run.
 - GKE Clusters (Google Kubernetes Engine).
 - Dataproc Clusters.
 - Cloud DNS managed zones and record sets.
 - Cloud Deployment Manager deployments.
- **Other CLI Tools for Specific GCP Services:**
 - Cloud Bigtable: `cbt`
 - Cloud BigQuery: `bq`
 - Cloud Storage: `gsutil` (older tool), `gcloud` is recommended for future use.
 - Kubernetes: `kubectl` (manages resources in Kubernetes clusters).

CODE

Step-02: Create VM using gcloud CLI

- **TAGS:** Tags allow network firewall rules and routes to be applied to specified VM instances
- **Important Note:** Upload `webserver-install.sh` to Google Cloud shell if running `gcloud` commands on cloud shell
- [gcloud Documentation](#)

```
# Set Project
gcloud config set project PROJECT_ID
gcloud config set project RevGCP

# Create VM Instance
gcloud compute instances create demo3-vm-gcloud \
  --zone=us-central1-a \
```

```

--machine-type=e2-micro \
--network-interface=subnet=default \
--tags=http-server \
--metadata-from-file=startup-script=webserver-install.sh
# upload the file

# List Compute Instances
gcloud compute instances list

## Optional Commands - For reference
# To list instances with their respective status and tags, run:
gcloud compute instances list --format='table(name,status,tags.list())'

# To list instances tagged with a specific tag, tag1, run:
gcloud compute instances list --filter='tags:http-server'

```

Step-03: Stop and Start a VM Instance

```

# Stop VM Instance
gcloud compute instances stop demo3-vm-gcloud --zone=us-central1-a
gcloud compute instances list --filter='name:demo3-vm-gcloud'

# Start VM Instance
gcloud compute instances start demo3-vm-gcloud --zone=us-central1-a
gcloud compute instances list --filter='name:demo3-vm-gcloud'

```

Step-04: Update VM using gcloud CLI

```

# Update VM: Enable deletion protection
gcloud compute instances update demo3-vm-gcloud \
  --zone=us-central1-a \
  --deletion-protection

# Delete VM
gcloud compute instances delete demo3-vm-gcloud --zone=us-central1-a

## ERROR MESSAGE
ERROR: (gcloud.compute.instances.delete) Could not fetch resource:
- Invalid resource usage: 'Resource cannot be deleted if it's protected against deletion.

# Update VM: Disable deletion protection
gcloud compute instances update demo3-vm-gcloud \
  --zone=us-central1-a \
  --no-deletion-protection

# Delete VM
gcloud compute instances delete demo3-vm-gcloud --zone=us-central1-a

```